MAPS OUT OF A LOCALISED MODULE ARE THE MAPS OUT OF THE MODULE, WHEN THE TARGET HAS INVERTIBLE ACTIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Description of a localised module by fractions. Used throughout the counted proof (lines 1400-1420). Lets an element of the localisation be written as m/s, which is what both injectivity and surjectivity are checked on.
- Invertibility of the actions of the elements of S on N. The hypothesis of the statement, used at lines 1401 and 1409. Allows the action of s to be undone when defining the extension and when showing the map is determined by its values on M.

Lemma 0.1. *Let R be a ring. Let $S \subset R$ a multiplicative subset. Let M, N be R-modules. Assume all the elements of S act as automorphisms on N. Then the canonical map*

$$\mathrm{Hom}_R(S^{-1}M, N) \longrightarrow \mathrm{Hom}_R(M, N)$$

induced by the localization map, is an isomorphism.

Proof. It is clear that the map is well-defined and R-linear. Injectivity: Let $\alpha \in \mathrm{Hom}_R(S^{-1}M, N)$ and take an arbitrary element $m/s \in S^{-1}M$. Then, since $s \cdot \alpha(m/s) = \alpha(m/1)$, we have $\alpha(m/s) = s^{-1}(\alpha(m/1))$, so α is completely determined by what it does on the image of M in $S^{-1}M$. Surjectivity: Let $\beta : M \to N$ be a

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given R-linear map. We need to show that it can be "extended" to $S^{-1}M$. Define a map of sets

$$M \times S \to N, \quad (m, s) \mapsto s^{-1}\beta(m)$$

Clearly, this map respects the equivalence relation from above, so it descends to a well-defined map $\alpha : S^{-1}M \to N$. It remains to show that this map is R-linear, so take $r, r' \in R$ as well as $s, s' \in S$ and $m, m' \in M$. Then

$$\begin{aligned}
\alpha(r \cdot m/s + r' \cdot m'/s') &= \alpha((r \cdot s' \cdot m + r' \cdot s \cdot m')/(ss')) \\
&= (ss')^{-1}\beta(r \cdot s' \cdot m + r' \cdot s \cdot m') \\
&= (ss')^{-1}(r \cdot s'\beta(m) + r' \cdot s\beta(m')) \\
&= r\alpha(m/s) + r'\alpha(m'/s')
\end{aligned}$$

and we win. □